

Experiment Report

Name: 李崇楷

Institution: NYCU AIIT

Course: Machine Learning

Student ID: 313834006

Code: <https://github.com/kailee0422/Machine-Learning.git>

Professor: Ching-Wen Ma

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Abstract

This study explores the mathematical foundation and application of Poisson regression in the context of Generalized Linear Models (GLMs). Poisson regression is commonly used to model count data, particularly when the response variable follows a Poisson distribution. We begin by demonstrating that the Poisson distribution belongs to the exponential family of distributions, identifying the key elements such as the canonical response function and sufficient statistics. Utilizing these properties, we derive the log-likelihood function for a Poisson GLM and apply stochastic gradient ascent to optimize the model parameters. Additionally, we investigate the convexity of the Negative Log-Likelihood (NLL) loss function in GLMs, proving that the Hessian matrix of the loss function is Positive Semi-Definite (PSD), thereby guaranteeing global optimality. This work provides both a theoretical understanding of GLM convexity and a practical framework for implementing Poisson regression to predict real-world traffic data, highlighting the broader applicability of exponential family distributions in machine learning models.

Object

Problem set 3 (a): **Show that Poisson Distribution is in the Exponential Family**

The objective of this problem is to express the Poisson distribution in the form of the exponential family. By doing so, we will identify key components such as the natural parameter (η), the sufficient statistic ($T(y)$), and other relevant terms. This exercise demonstrates the connection between the Poisson distribution and the broader exponential family, a foundational concept in Generalized Linear Models (GLMs).

Problem set 3 (b): **Identify the Canonical Response Function for Poisson Regression**

The objective of this problem is to derive the canonical response function for the Poisson family in a GLM context. This involves determining the link function that relates the expected value of the Poisson-distributed response variable to the linear predictor, providing a key insight into the relationship between the input features and the model's predicted outcome.

Problem set 3 (c): **Derive the Stochastic Gradient Ascent Rule for Poisson Regression**

The objective of this problem is to derive the update rule for stochastic gradient ascent in the context of Poisson regression. By differentiating the log-likelihood of the Poisson model with respect to the model parameters, we will obtain an expression that can be used to iteratively update the parameters during model training.

Problem set 3 (d): **Implement Poisson Regression to Predict Website Traffic**

The objective of this problem is to implement Poisson regression using gradient ascent to predict daily website traffic. We will work with a real-world dataset and develop a model that assumes the number of visitors follows a Poisson distribution. The goal is to maximize the log-likelihood of the parameters and evaluate the model's performance on the provided dataset.

Problem set 4 (a): Derive the Mean of the Exponential Family Distribution

The objective of this problem is to derive an expression for the mean of an exponential family distribution in terms of the natural parameter η . Specifically, we will show that the expected value of the response variable Y , conditioned on the input features, can be written as the gradient of the log-partition function with respect to η . This highlights the connection between the log-partition function and the distribution's mean, a fundamental property of the exponential family.

Problem set 4 (b): Derive the Variance of the Exponential Family Distribution

The objective of this problem is to derive an expression for the variance of the exponential family distribution. We will demonstrate that the variance can be expressed as the derivative of the mean with respect to η , or equivalently, the second derivative of the log-partition function. This result further underscores the mathematical convenience of the exponential family, where key distributional properties like variance can be obtained through derivatives.

Problem set 4 (c): Prove the Convexity of the Negative Log-Likelihood for GLMs

The objective of this problem is to prove that the negative log-likelihood (NLL) loss function for a Generalized Linear Model (GLM) is convex with respect to the model parameters. To do this, we will first write out the NLL loss as a function of the parameters, then calculate the Hessian of the loss. By showing that the Hessian is always positive semi-definite (PSD), we will conclude that the NLL loss is convex, ensuring that any local minimum is also a global minimum.

Problem Methods and Result

Problem set 3(a):

Problem:

(a) [5 points] Consider the Poisson distribution parameterized by λ :

$$p(y; \lambda) = \frac{e^{-\lambda} \lambda^y}{y!}.$$

Show that the Poisson distribution is in the exponential family, and clearly state the values for $b(y)$, η , $T(y)$, and $a(\eta)$.

Solution and Result:

$$\begin{aligned} p(y; \lambda) &= \frac{e^{-\lambda} \lambda^y}{y!} = \frac{1}{y!} \exp(y \cdot \log \lambda - \lambda) \\ \text{Exponential Family } p(y; \eta) &= b(y) \cdot \exp(\eta \cdot T(y) - a(\eta)) \end{aligned} \quad \rightarrow \quad \begin{cases} b(y) = \frac{1}{y!} \\ \eta = \log \lambda \\ T(y) = y \\ a(\eta) = \lambda = e^\eta \\ (e^\eta = e^{\log \lambda} = \lambda) \end{cases} \quad \#$$

Problem set 3 (b):

Problem:

- (b) [3 points] Consider performing regression using a GLM model with a Poisson response variable. What is the canonical response function for the family? (You may use the fact that a Poisson random variable with parameter λ has mean λ .)

Solution and Result:

Poisson-distributed random y

$$E(y|x;\theta) = \lambda, \quad \eta = \theta^T x$$

$$\Rightarrow \lambda = e^\eta = e^{\theta^T x} \quad \Rightarrow \underline{h_\theta(x) = E(y|x;\theta) = \lambda = e^\eta = e^{\theta^T x}} \quad \#$$

(Due to exponential family)

Problem set 3 (c):

Problem:

- (c) [7 points] For a training set $\{(x^{(i)}, y^{(i)}); i = 1, \dots, m\}$, let the log-likelihood of an example be $\log p(y^{(i)}|x^{(i)}; \theta)$. By taking the derivative of the log-likelihood with respect to θ_j , derive the stochastic gradient ascent update rule for learning using a GLM model with Poisson responses y and the canonical response function.

Solution and Result:

(c)

$$\log p(y^{(i)}|x^{(i)}; \theta) =$$

$$\Rightarrow \log \left(\frac{1}{y^{(i)}!} \cdot \exp(\underbrace{\theta^T x^{(i)}}_{\eta} \cdot \underbrace{y^{(i)}}_{y} - \underbrace{e^{\theta^T x^{(i)}}}_{e^\eta}) \right) \stackrel{\text{Simplify}}{=} \log p(y^{(i)}|x^{(i)}; \theta) = -\log y^{(i)}! + \theta^T x^{(i)} y^{(i)} - e^{\theta^T x^{(i)}}$$

To update parameter θ , we need the gradient of the log-likelihood with respect to θ_j

$$\text{Derivative } \theta_j \Rightarrow \frac{\partial \log p(y^{(i)}|x^{(i)}; \theta)}{\partial \theta_j} = y^{(i)} \cdot x_j^{(i)} - e^{\theta^T x^{(i)}} \cdot x_j^{(i)} = (y^{(i)} - e^{\theta^T x^{(i)}}) \cdot x_j^{(i)}$$

Stochastic Gradient
Ascent

$$\Rightarrow \underline{\theta_j := \theta_j + \alpha (y^{(i)} - e^{\theta^T x^{(i)}}) \cdot x_j^{(i)}} \quad \#$$

$e^{\theta^T x^{(i)}}$ is expected value of Poisson Distribution based on $\theta, x^{(i)}$
 α is learning rate

Problem set 3 (d):

Problem:

- (d) [7 points] **Coding problem.** Consider a website that wants to predict its daily traffic. The website owners have collected a dataset of past traffic to their website, along with some features which they think are useful in predicting the number of visitors per day. The dataset is split into train/valid/test sets and follows the same format as Datasets 1-3:

data/ds4_{train,valid}.csv

We will apply Poisson regression to model the number of visitors per day. Note that applying Poisson regression in particular assumes that the data follows a Poisson distribution whose natural parameter is a linear combination of the input features (*i.e.*, $\eta = \theta^T x$). In `src/p03d_poisson.py`, implement Poisson regression for this dataset and use gradient ascent to maximize the log-likelihood of θ .

Solution:

```
import matplotlib.pyplot as plt
import numpy as np
import util

from linear_model import LinearModel

x_train, y_train = util.load_dataset('ds4_train.csv', add_intercept=True)
x_valid, y_valid = util.load_dataset('ds4_valid.csv', add_intercept=True)
```

Code: Import and Load train/valid data

- `matplotlib.pyplot` is used for plotting.
- `numpy` provides matrix operations.
- `util.load_dataset()` loads the dataset.
- `LinearModel` is a base class for linear models.

```
def h(self, theta, x):
    """Vectorized implementation of h.

    :param theta: Shape (n,).
    :param x:      Training example inputs. Shape (m, n).
    :return:        The hypothesis of Poisson Regression, given by  $h(x) = \exp(\theta^T x)$ . Shape (m,).
    """
    return np.exp(x @ theta)
```

Code: This function calculates $h(x)=\exp(\theta^T x)$, the hypothesis for Poisson regression, which estimates the expected number of visitors per day.

```
def fit(self, x, y):
    """Run gradient ascent to maximize likelihood for Poisson regression.

    :param x: Training example inputs. Shape (m, n).
    :param y: Training example labels. Shape (m,).
    """

    def next_step(theta):
        return self.step_size / m * x.T @ (y - self.h(theta, x))

    m, n = x.shape

    # Initialize theta
    if self.theta is None:
        theta = np.zeros(n)
    else:
        theta = self.theta

    # Update theta
    step = next_step(theta)
    while np.linalg.norm(step, 1) >= self.eps:
        theta += step
        step = next_step(theta)

    self.theta = theta
```

Code: **Implements gradient ascent to maximize the log-likelihood**

- Computes the gradient as $\frac{1}{m}X^T(y-h(x))$, where y is the actual count of visitors and $h(x)$ is the predicted count.
- The loop updates theta using the computed gradient until the change in theta (step size) becomes smaller than a threshold (self.eps).

```
def predict(self, x):
    """Make a prediction given inputs x.

    :param x: Inputs of shape (m, n).
    :return: Floating-point prediction for each input, shape (m,).
    """

    return self.h(self.theta, x)
```

Code: **Predicts the number of visitors based on the features x using the learned theta values.**

```
clf = PoissonRegression(step_size=2e-7)
clf.fit(x_train, y_train)
```

Code: **The model is trained with a small learning rate (2e-7) on the training set, which controls how large each step of gradient ascent is.**

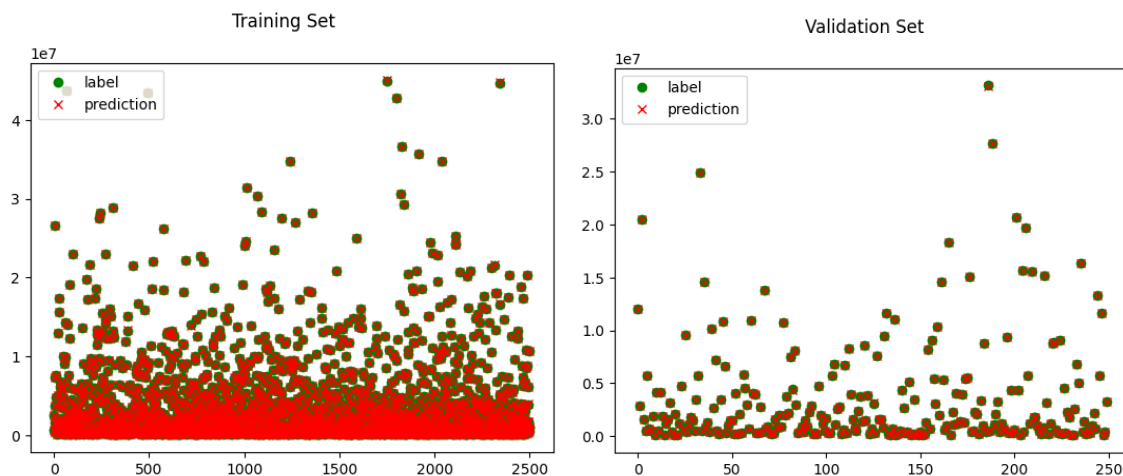
```
def plot(y_label, y_pred, title):
    plt.plot(y_label, 'go', label='label')
    plt.plot(y_pred, 'rx', label='prediction')
    plt.suptitle(title, fontsize=12)
    plt.legend(loc='upper left')
```

```
y_train_pred = clf.predict(x_train)
plot(y_train, y_train_pred, 'Training Set')
```

```
y_valid_pred = clf.predict(x_valid)
plot(y_valid, y_valid_pred, 'Validation Set')
```

Code: **Plot the predictions on training and validation set**

Result



Problem set 4:

4. [15 points] Convexity of Generalized Linear Models

In this question we will explore and show some nice properties of Generalized Linear Models, specifically those related to its use of Exponential Family distributions to model the output.

Most commonly, GLMs are trained by using the negative log-likelihood (NLL) as the loss function. This is mathematically equivalent to Maximum Likelihood Estimation (*i.e.*, maximizing the log-likelihood is equivalent to minimizing the negative log-likelihood). In this problem, our goal is to show that the NLL loss of a GLM is a convex function w.r.t the model parameters. As a reminder, this is convenient because a convex function is one for which any local minimum is also a global minimum.

To recap, an exponential family distribution is one whose probability density can be represented

$$p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta)),$$

where η is the *natural parameter* of the distribution. Moreover, in a Generalized Linear Model, η is modeled as $\theta^T x$, where $x \in \mathbb{R}^n$ are the input features of the example, and $\theta \in \mathbb{R}^n$ are learnable parameters. In order to show that the NLL loss is convex for GLMs, we break down the process into sub-parts, and approach them one at a time. Our approach is to show that the second derivative (*i.e.*, Hessian) of the loss w.r.t the model parameters is Positive Semi-Definite (PSD) at all values of the model parameters. We will also show some nice properties of Exponential Family distributions as intermediate steps.

For the sake of convenience we restrict ourselves to the case where η is a scalar. Assume $p(Y|X; \theta) \sim \text{ExponentialFamily}(\eta)$, where $\eta \in \mathbb{R}$ is a scalar, and $T(y) = y$. This makes the exponential family representation take the form

$$p(y; \eta) = b(y) \exp(\eta y - a(\eta)).$$

Problem set 4(a+b):

Problem:

- (a) [5 points] Derive an expression for the mean of the distribution. Show that $\mathbb{E}[Y | X; \theta]$ can be represented as the gradient of the log-partition function a with respect to the natural parameter η .

Hint: Start with observing that $\frac{\partial}{\partial \eta} \int p(y; \eta) dy = \int \frac{\partial}{\partial \eta} p(y; \eta) dy$.

- (b) [5 points] Next, derive an expression for the variance of the distribution. In particular, show that $\text{Var}(Y | X; \theta)$ can be expressed as the derivative of the mean w.r.t η (i.e., the second derivative of the log-partition function $a(\eta)$ w.r.t the natural parameter η).

Solution and Result:

(a)

$$\frac{\partial}{\partial \eta} \int p(y; \eta) dy = 0 \quad (p(y; \eta) = 1)$$

Apply Exponential family form

$$\frac{\partial}{\partial \eta} \int p(y; \eta) dy = \int \frac{\partial}{\partial \eta} p(y; \eta) dy \quad p(y; \eta) = b(y) \cdot \exp(\eta \cdot y - a(\eta))$$

$$\frac{\partial}{\partial \eta} p(y; \eta) = p(y; \eta) \cdot \left(y - \frac{\partial a(\eta)}{\partial \eta} \right)$$

$$\int p(y; \eta) \left(y - \frac{\partial a(\eta)}{\partial \eta} \right) dy = \int p(y; \eta) \cdot y \cdot dy - \frac{\partial a(\eta)}{\partial \eta} \int p(y; \eta) dy = 0$$

$$\left(\text{Since } \int p(y; \eta) dy = 1 \right) = \mathbb{E}[Y; \eta] - \frac{\partial a(\eta)}{\partial \eta} = 0$$

$$\Rightarrow \mathbb{E}[Y; \eta] = \frac{\partial a(\eta)}{\partial \eta} \quad \#$$

(b)

$$\mathbb{E}[Y] = \int y \cdot p(y; \eta) dy$$

expected value of Y

$$\frac{\partial \mathbb{E}[Y]}{\partial \eta} = \frac{\partial}{\partial \eta} \int y \cdot p(y; \eta) dy \stackrel{\text{Leibniz rule}}{=} \int y \cdot \frac{\partial}{\partial \eta} p(y; \eta) dy$$

by (a) calculate result

$$\frac{\partial}{\partial \eta} p(y; \eta) = p(y; \eta) \cdot \left(y - \frac{\partial a(\eta)}{\partial \eta} \right)$$

$$\Rightarrow \int y \cdot p(y; \eta) \cdot \left(y - \frac{\partial a(\eta)}{\partial \eta} \right) dy = \int y^2 p(y; \eta) dy - \frac{\partial a(\eta)}{\partial \eta} \int y p(y; \eta) dy$$

$$= \mathbb{E}[Y^2] - \frac{\partial a(\eta)}{\partial \eta} \cdot \mathbb{E}[Y]$$

$$\stackrel{\text{by (a)}}{=} \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 = \frac{\partial}{\partial \eta} \mathbb{E}[Y] = \text{Var}(Y) \quad (= \mathbb{E}[Y^2] - \mathbb{E}[Y]^2)$$

$$\mathbb{E}[Y] = \frac{\partial a(\eta)}{\partial \eta} \quad \Rightarrow \quad \text{Var}(Y) = \frac{\partial^2 a(\eta)}{\partial^2 \eta} \quad \#$$

Problem set 4(c):

Problem:

- (c) [5 points] Finally, write out the loss function $\ell(\theta)$, the NLL of the distribution, as a function of θ . Then, calculate the Hessian of the loss w.r.t θ , and show that it is always PSD. This concludes the proof that NLL loss of GLM is convex.

Hint: Use the chain rule of calculus along with the results of the previous parts to simplify your derivations.

Solution and Result:

(c)

$$\ell(\theta) = -\sum_{i=1}^m \log p(y^{(i)} | x^{(i)}; \theta) = -\sum_{i=1}^m (\log b(y^{(i)}) + \eta^{(i)} \cdot y^{(i)} - a(\eta^{(i)})) = -\sum_{i=1}^m (\theta^T x^{(i)} \cdot y^{(i)} - a(\theta^T x^{(i)}))$$

(1) exponential family
log b don't depend on θ , so can ignore.
 $\eta^{(i)} = \theta^T x^{(i)}$

$$\frac{\partial \ell(\theta)}{\partial \theta_j} = \sum_{i=1}^m (\underbrace{a'(\theta^T x^{(i)})}_{\text{by (a)}} \cdot x_j^{(i)} - y^{(i)} x_j^{(i)}) = \sum_{i=1}^m (E[Y^{(i)} | X^{(i)}; \theta] \cdot x_j^{(i)} - y^{(i)} x_j^{(i)})$$

$$H_{jk} = \frac{\partial^2 \ell(\theta)}{\partial \theta_j \partial \theta_k} = \sum_{i=1}^m \underbrace{a''(\theta^T x^{(i)})}_{\text{by (b)}} \cdot x_j^{(i)} \cdot x_k^{(i)} = \sum_{i=1}^m \text{Var}(Y^{(i)} | X^{(i)}; \theta) \cdot x_j^{(i)} \cdot x_k^{(i)}$$

To show H is positive definite, $z \in \mathbb{R}^n$, and let $z^T H \cdot z$ is non-negative

$$z^T H \cdot z = \sum_{i=1}^m a''(\theta^T x^{(i)}) \left(\sum_{j=1}^n x_j^{(i)} z_j \right)^2 = \sum_{i=1}^m \text{Var}(Y | X^{(i)}; \theta) \cdot ((x^{(i)})^T \cdot z)^2$$

and since $\text{Var}(Y | X^{(i)}; \theta) \geq 0$ and $(\)^2 \geq 0$, $\Rightarrow z^T H \cdot z \geq 0 \Rightarrow$ Positive semi-Definite

Any local min is global m.
 $\Uparrow$
 NLL loss function is convex with respect to θ
 $\Uparrow$

References

1. Maxim5. (2018). CS229 2018 Autumn Problem Sets and Solutions. GitHub. Retrieved from <https://github.com/maxim5/cs229-2018-autumn/tree/main/problem-sets-solutions/PS1>